

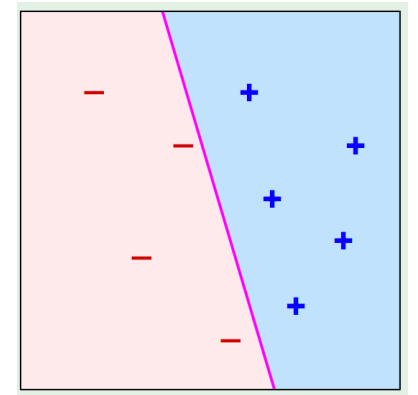
Linear SVM

Support Vector Machines (SVM)

Kernel SVM

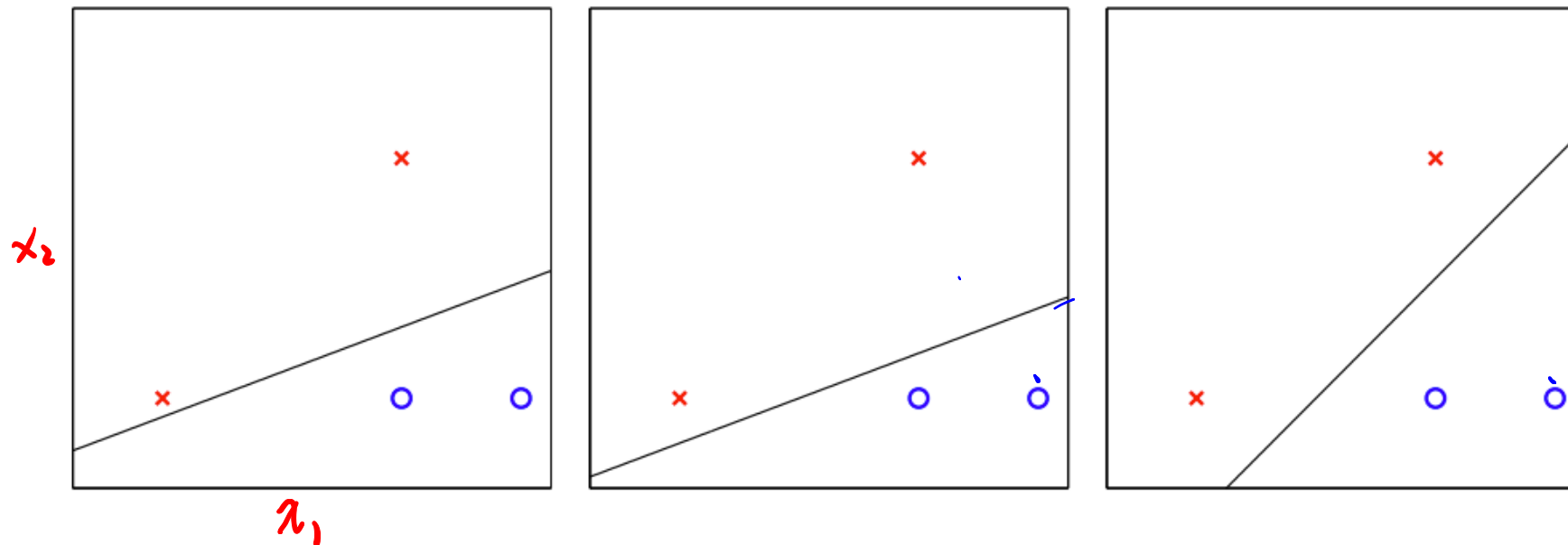
What Do We Know about Linear Classification?

- What we discussed so far:
 - PLA: Find a linear separator that separates the data within finite steps, if data is linear separable.
 - Pocket algorithm: empirically keep the best separator during PLA.
 - Surrogate loss: Using logistic regression for linear classification.
- Challenges
 - Binary classification error is hard to optimize,
 - We cannot use “gradient descent” type of algorithm minimize E_{in} .
- Support vector machines (SVM) tries to look at things a bit differently.



Linear Classification

- Which separator would you choose?



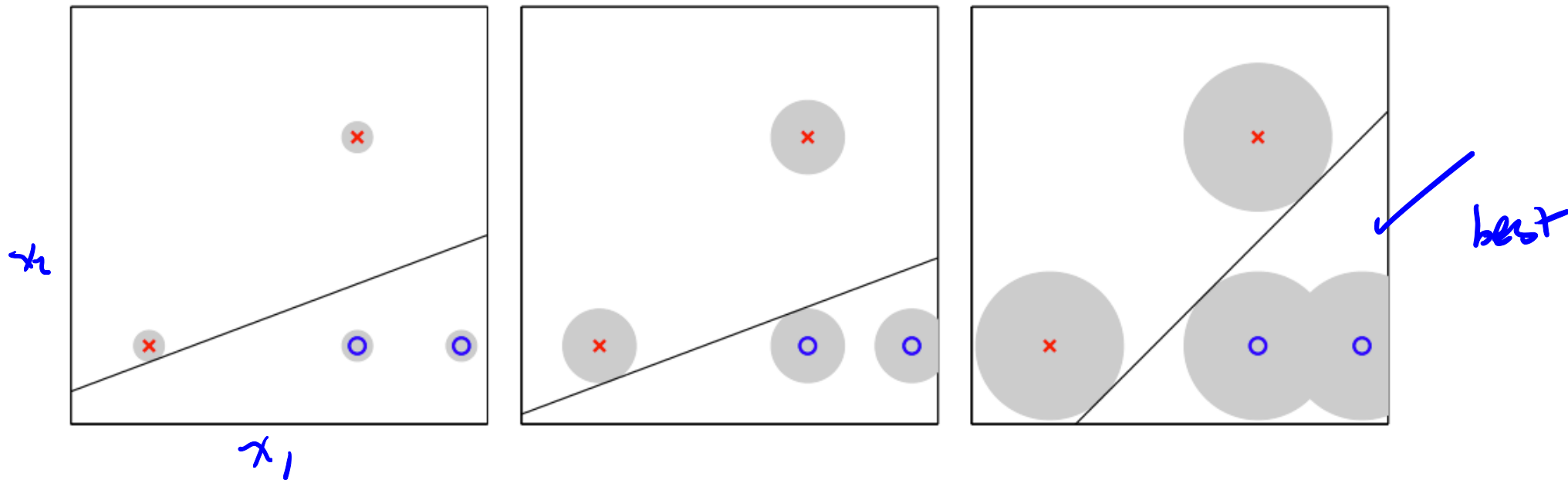
Probably the right one.
Why?

Linear Classification

measurement error
data + noise

- Which separator would you choose?

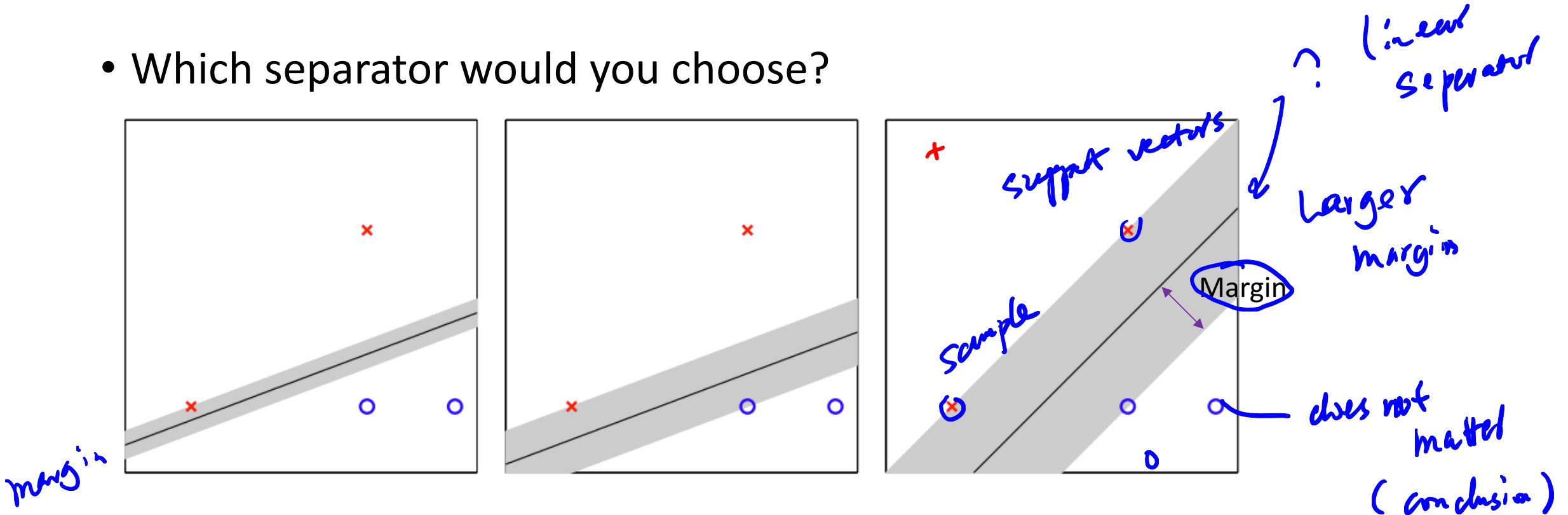
intuitively



More robust to noise (e.g., measurement error of $\vec{x}$)

Linear Classification

- Which separator would you choose?

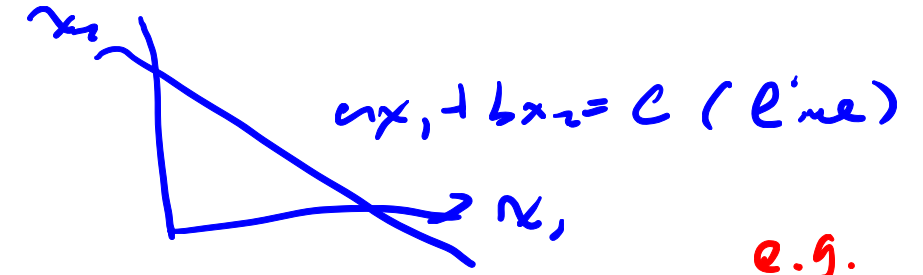


Margin: shortest distance from the separator to the points in D
(Informal argument)

Higher margin $\Rightarrow$ more "constrained" hypothesis $\Rightarrow$ lower VC dimension

Handwritten blue notes: "stronger classifier model" is written above "constrained", and "VC" is circled in blue.

Support Vector Machine



- Goal:
 - Find the **max-margin** linear separator that separates the data
 - Linear separator $w_i^T \vec{x} + w_0 = 0$ (line)

Notations:

Notations we used so far:

- $\vec{x} = (x_0, x_1, \dots, x_d)$
- $\vec{w} = (w_0, w_1, \dots, w_d)$
- Linear separator

$$h(\vec{x}) = \text{sign}(\vec{w}^T \vec{x})$$

Notations we will use in SVM

- $\vec{x} = (x_1, \dots, x_d)$
- $\vec{w} = (w_1, \dots, w_d)$
- Linear separator

$$h(\vec{x}) = \text{sign}(\vec{w}^T \vec{x} + \underline{b})$$

$$\begin{aligned} w^T \vec{x} &\geq 0 \\ w^T \vec{x} &\leq 0 \end{aligned}$$

$$\begin{aligned} w^T \vec{x} &\geq 1 \\ w^T \vec{x} &\leq -1 \end{aligned}$$

$\oplus$
 $\ominus$

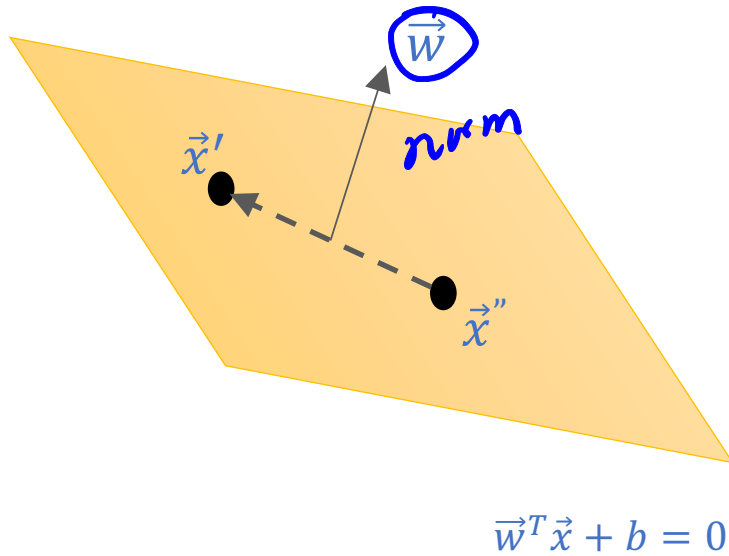
Harder constraints

Separating the bias/intercept $\underline{b}$ is important for us to characterize the margin.

We will use $(\vec{w}, b)$ to characterize the hypothesis

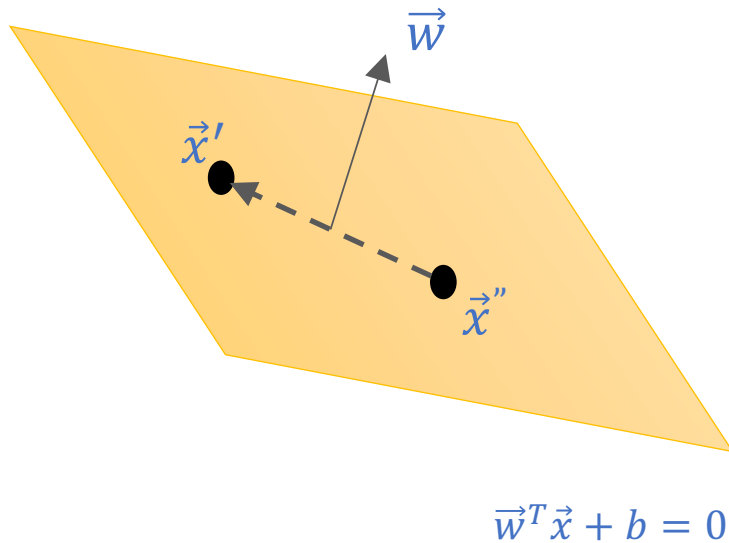
Relevant Review of Linear Algebra

- Claim: $\vec{w}$ is the norm vector of the hyperplane $\vec{w}^T \vec{x} + b = 0$



Relevant Review of Linear Algebra

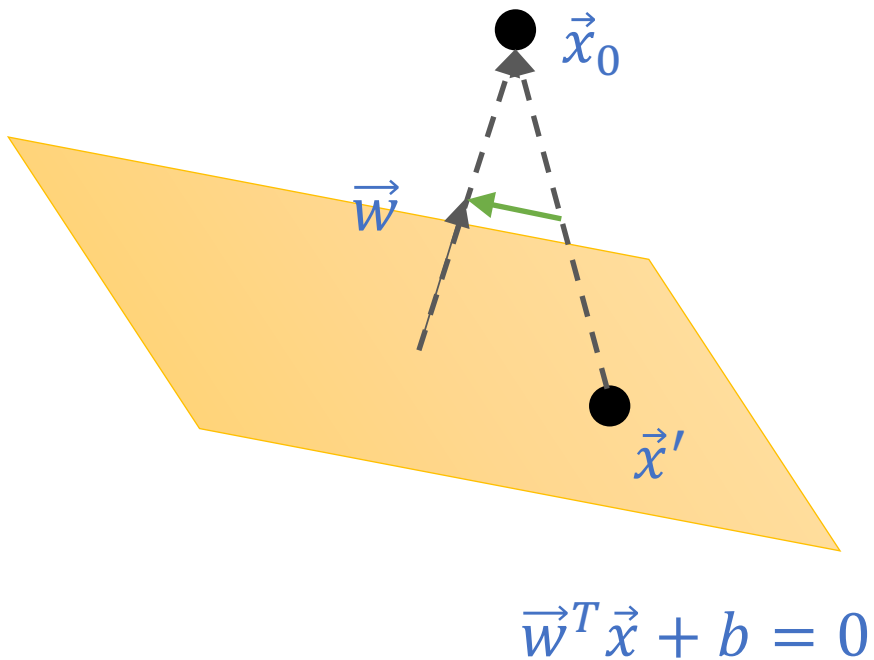
- Claim: $\vec{w}$ is the norm vector of the hyperplane $\vec{w}^T \vec{x} + b = 0$



- Consider any two points $\vec{x}'$ and $\vec{x}''$ on the hyperplane
 - $\vec{w}^T \vec{x}' + b = 0$
 - $\vec{w}^T \vec{x}'' + b = 0$
- Combining the above
 - $\vec{w}^T (\vec{x}' - \vec{x}'') = 0$ *orthogonal*
- $\vec{w}$ is orthogonal to the hyperplane
- $\vec{w}$ is the norm vector of the hyperplane

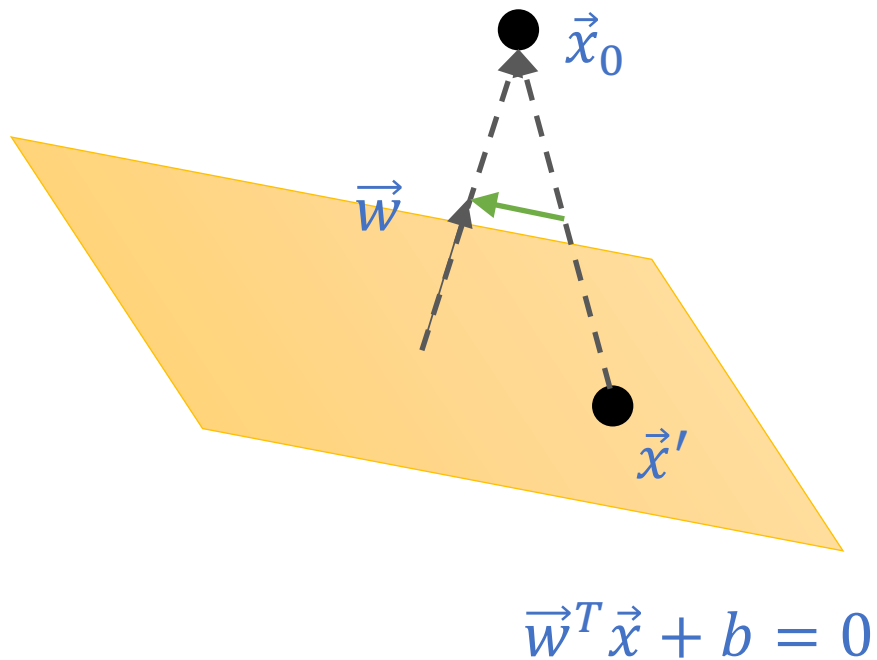
Relevant Review of Linear Algebra

- What is the distance between a point $\vec{x}_0$ and a hyperplane $\vec{w}^T \vec{x} + b = 0$



Relevant Review of Linear Algebra

- What is the distance between a point $\vec{x}_0$ and a hyperplane $\vec{w}^T \vec{x} + b = 0$



- Consider an arbitrary point $\vec{x}'$ on the hyperplane
- Distance between the point $\vec{x}$ and the hyperplane

$$\begin{aligned} \text{dist}(\vec{x}_0, \vec{w}, b) &= \left| \frac{\vec{w}^T}{\|\vec{w}\|} (\vec{x}_0 - \vec{x}') \right| \\ &= \left| \frac{1}{\|\vec{w}\|} (\vec{w}^T \vec{x}_0 - \vec{w}^T \vec{x}') \right| \\ &= \left| \frac{1}{\|\vec{w}\|} (\vec{w}^T \vec{x}_0 + b) \right| \end{aligned}$$

Outline of Our Discussion

- Assume data is linearly separable

- Formulate the **hard-margin SVM**

linear SVM

soft margin
(outliers)

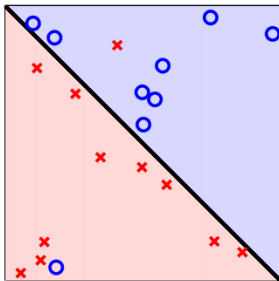
Given D , find separator $(\vec{w}, b)$ that
maximize $\text{margin}(\vec{w}, b)$
s.t. all points in D is correctly classified

margin small
large

- When data is not linearly separable

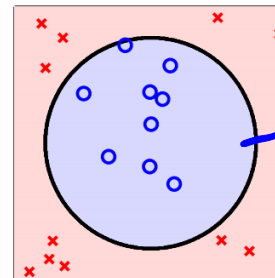
- Tolerate some noise

- **Soft-margin SVM**



- Nonlinear transform

- **Dual formulation and kernel tricks**



non linear SVM

Hard-Margin SVM

Hard-Margin SVM

$$\begin{aligned} y_n &= +1 \\ y_n &= -1 \end{aligned}$$

$$\begin{aligned} \vec{w}^T \vec{x}_n + b &> 0 \\ \vec{w}^T \vec{x}_n + b &< 0 \end{aligned}$$

- Goal
 - Given $D = \{(\vec{x}_1, y_1), \dots, (\vec{x}_N, y_N)\}$ that is **linearly separable**
 - Find separator $(\vec{w}, b)$ that (1) maximizes the margin and (2) separates D
- $(\vec{w}, b)$ separates D (making correct predictions for all points in D)
 - $y_n = \text{sign}(\vec{w}^T \vec{x}_n + b)$ for all n $\vec{x}_n, y_n \in \{+1, -1\}$
 - $y_n(\vec{w}^T \vec{x}_n + b) \geq 0$ for all n Correct prediction "constraint"
- Margin: shortest distance from the separator to points in D

$$\begin{aligned} \text{margin}(\vec{w}, b) &= \min_n \text{dist}(\vec{x}_n, \vec{w}, b) \\ &= \min_n \left| \frac{1}{\|\vec{w}\|} (\vec{w}^T \vec{x}_n + b) \right| \\ \text{margin} &= \min_n \frac{1}{\|\vec{w}\|} y_n (\vec{w}^T \vec{x}_n + b) \end{aligned}$$

$$\text{dist}(\vec{x}_0, \vec{w}, b) = \left| \frac{1}{\|\vec{w}\|} (\vec{w}^T \vec{x}_0 - b) \right|$$

$$\begin{aligned} y_n &\in \{-1, +1\} \text{ and} \\ y_n(\vec{w}^T \vec{x}_n + b) &\geq 0 \end{aligned}$$

Hard-Margin SVM

- Goal
 - Given $D = \{(\vec{x}_1, y_1), \dots, (\vec{x}_N, y_N)\}$ that is **linearly separable**
 - Find separator $(\vec{w}, b)$ that (1) maximizes the margin and (2) separates D
- Formulate it as a constrained optimization problem

$$\begin{array}{ll} \text{maximize}_{\vec{w}, b} & \text{margin}(\vec{w}, b) \\ \text{subject to} & \underline{y_n(\vec{w}^T \vec{x}_n + b) \geq 0, \forall n} \end{array}$$

separable

$$\text{margin}(\vec{w}, b) = \min_n \frac{1}{\|\vec{w}\|} y_n(\vec{w}^T \vec{x}_n + b)$$

Hard-Margin SVM

- The constrained optimization problem

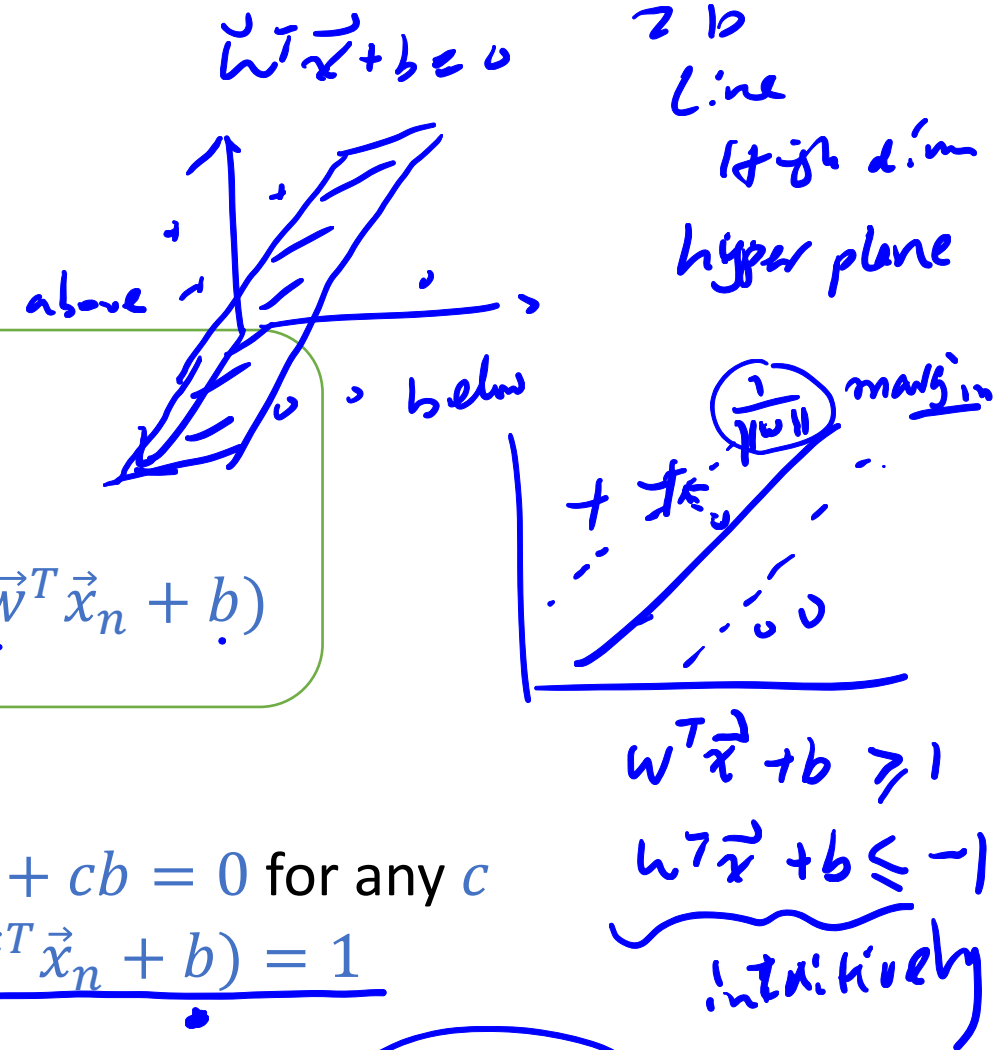
$$\begin{aligned} & \text{maximize}_{\vec{w}, b} \quad \text{margin}(\vec{w}, b) \\ & \text{subject to} \quad y_n(\vec{w}^T \vec{x}_n + b) \geq 0, \forall n \\ & \quad \quad \quad \text{margin}(\vec{w}, b) = \min_n \frac{1}{\|\vec{w}\|} y_n(\vec{w}^T \vec{x}_n + b) \end{aligned}$$

- normalizing $(\vec{w}, b)$

trick

- Note that $\vec{w}^T \vec{x} + b = 0$ is equivalent to $c\vec{w}^T \vec{x} + cb = 0$ for any c
- We will normalize $(\vec{w}, b)$ such that $\min_n y_n(\vec{w}^T \vec{x}_n + b) = 1$
 - $y_n(\vec{w}^T \vec{x}_n + b) \geq 1, \forall n$

$$\text{margin}(\vec{w}, b) = \frac{1}{\|\vec{w}\|}$$



$$|\vec{w}^T \vec{x} + b| \geq 1$$

Hard-Margin SVM

$\vec{w}$. b

- The constrained optimization problem

$$\begin{array}{ll} \text{maximize}_{\vec{w}, b} & \frac{1}{\|\vec{w}\|} \\ \text{subject to} & y_n(\vec{w}^T \vec{x}_n + b) \geq 1, \forall n \end{array}$$

- Some final adjustments

$$\begin{array}{ll} \text{minimize}_{\vec{w}, b} & \frac{1}{2} \vec{w}^T \vec{w} \quad \text{Quadratic} \\ \text{subject to} & y_n(\vec{w}^T \vec{x}_n + b) \geq 1, \forall n \end{array}$$

problem

constraint

Final Form of Hard-Margin SVM

$$\begin{aligned} &\text{minimize}_{\vec{w}, b} \quad \frac{1}{2} \vec{w}^T \vec{w} \\ &\text{subject to} \quad \underline{y_n (\vec{w}^T \vec{x}_n + b)} \geq 1, \forall n \end{aligned}$$

$$w = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

- How to solve it?
 - Hard-margin SVM is a Quadratic Program
 - Standard form of Quadratic Program (QP)

$$\begin{aligned} &\text{minimize}_{\vec{u}} \quad \frac{1}{2} \vec{u}^T \underline{Q} \vec{u} + \vec{p}^T \vec{u} \\ &\text{subject to} \quad \underline{A} \vec{u} \geq \vec{c} \end{aligned}$$

$$u = w, \quad Q = I, \quad p = 0.$$

using QP

matlab solver

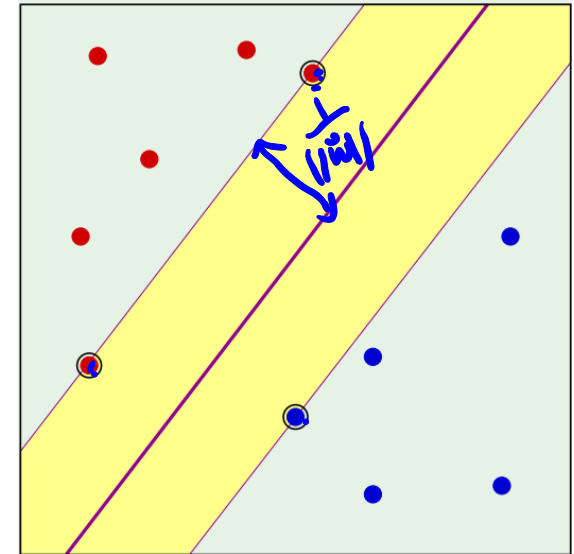
- There exist efficient QP solvers we can utilize

Support Vectors

positive $y_n = +1$, $\vec{w}^T \vec{x}_n + b = 1$
negative $y_n = -1$, $\vec{w}^T \vec{x}_n + b = -1$

- We call the points closest to the separator (candidate) **support vectors**
 - Since they **support** the separator
- What are the math properties of support vectors?
 - They are the points that the equality holds in the constraints
 - If $\vec{x}_n$ is a support vector, $y_n(\vec{w}^T \vec{x}_n + b) = 1$

$$\begin{array}{ll} \text{minimize}_{\vec{w}, b} & \frac{1}{2} \vec{w}^T \vec{w} \\ \text{subject to} & y_n(\vec{w}^T \vec{x}_n + b) \geq 1, \forall n \end{array}$$



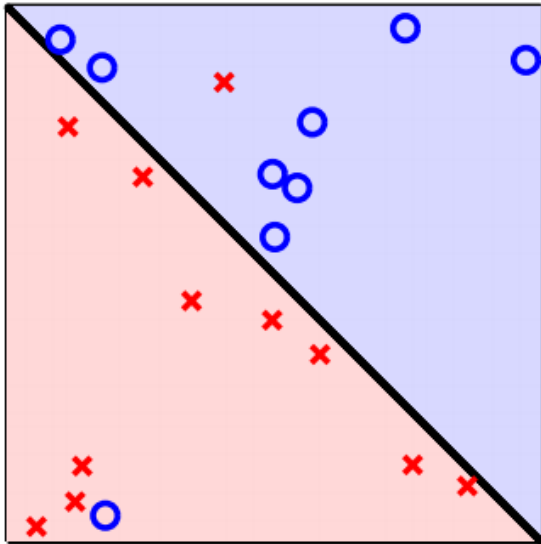
- Removing the non-support vectors will not impact the linear separator

What if Data is Not Linearly Separable

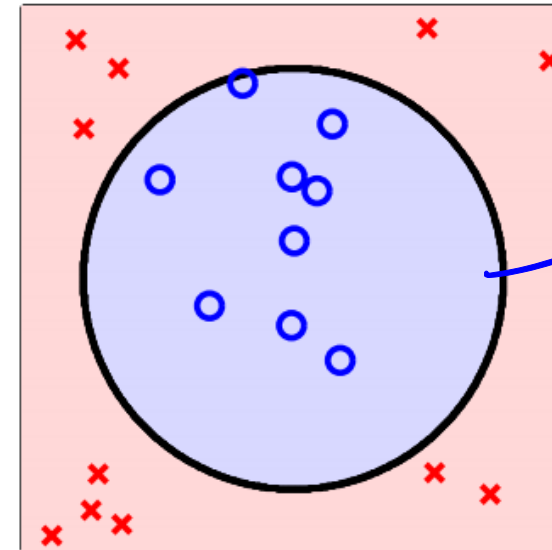
not perfectly linear separable. | Hard-margin

Non-Separable Data

- Two scenarios



- Tolerate some noise
 - Soft-Margin SVM



$(x-x_0)^2 + (y-y_0)^2 = C^2$
| kernel
trick
(projection)

- Nonlinear transform
 - Dual formulation and kernel tricks

Soft-Margin SVM

$$\max \frac{1}{\|\vec{w}\|}$$

- Intuition: We want to tolerate small noises when maintaining large margin

- For each point $(\vec{x}_n, y_n)$, we allow a deviation $\xi_n \geq 0$

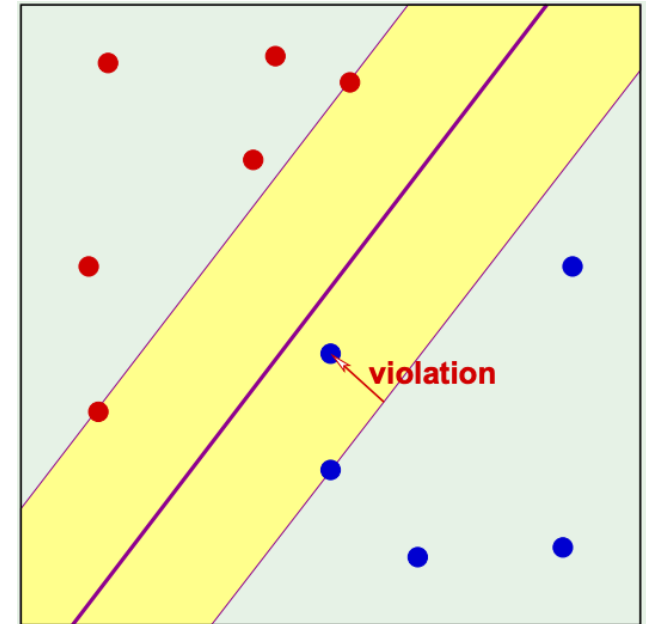
- Instead of requiring $y_n(\vec{w}^T \vec{x}_n + b) \geq 1$
- The constraint becomes

$$y_n(\vec{w}^T \vec{x}_n + b) \geq 1 - \xi_n$$

- We add a penalty for each deviation

- Total penalty $C \sum_{n=1}^N \xi_n$ deviation

constant weight

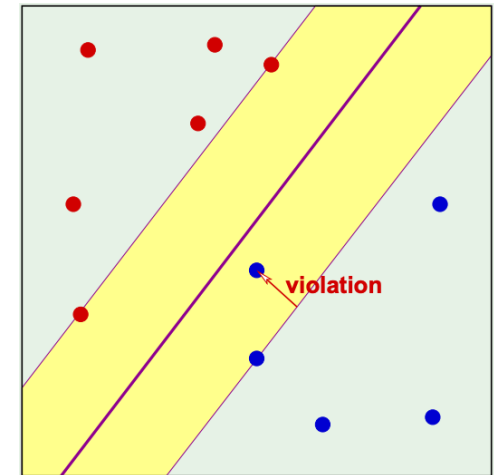


Soft-Margin SVM

- The constraint becomes: $y_n(\vec{w}^T \vec{x}_n + b) \geq 1 - \xi_n$
- We add a penalty for each deviation: Total penalty $C \sum_{n=1}^N \xi_n$

$$\begin{aligned} &\text{minimize}_{\vec{w}, b, \vec{\xi}} \quad \frac{1}{2} \vec{w}^T \vec{w} + C \sum_{n=1}^N \xi_n \\ &\text{subject to} \quad y_n(\vec{w}^T \vec{x}_n + b) \geq 1 - \xi_n, \forall n \\ &\quad \quad \quad \xi_n \geq 0, \forall n \end{aligned}$$

Penalty



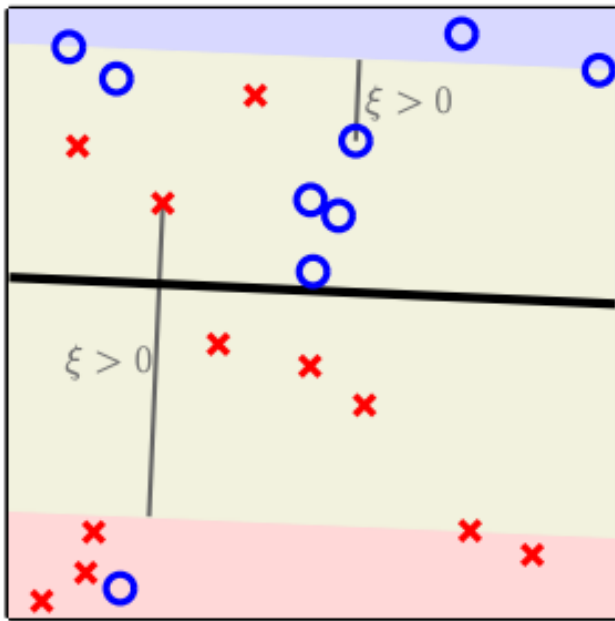
Remarks:

- C is a hyper-parameter we can choose, e.g., using validation
- Soft-margin SVM is still a Quadratic Program, with efficient solvers

Impacts of C in Soft-Margin SVM *linear svm*

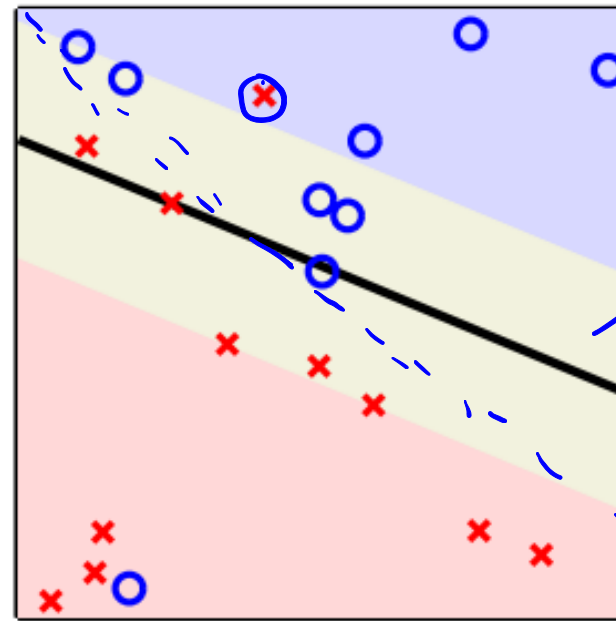
- Large C => less tolerate to noise, having smaller margin

larger margin



$C = 1$

tolerant noise

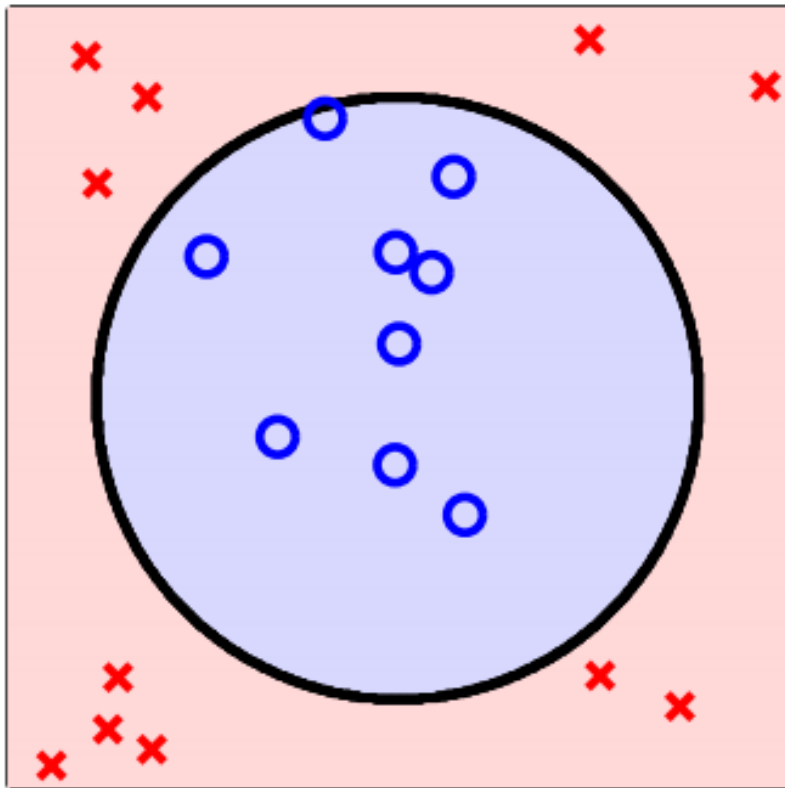


$C = 500$

*margin smaller
not good enough*

Nonlinear SVM

What if Tolerating Small Noises Is Not Enough



Nonlinear transform

We can apply standard nonlinear transformation procedure we talked about before

In SVM, we can combine the ideas of dual formulation and kernel tricks for the transformation

This is one of the key ingredients that makes SVM powerful